

Compliance Pack Accelerator

Databricks-native multi-jurisdiction compliance — pluggable regulation packs (DPDP India 2023 · UK GDPR · EU GDPR · CCPA · PIPEDA · ...)

ADR-0001 · Per-data-subject routing

Architectural foundation. Compliance is per-data-subject, not per-deployment. Every customer-level row carries a jurisdiction column; rules apply per principal through the regulation pack governing them. An Indian SaaS with UK clients applies DPDP rules to Indian principals (730-day retention) and UK GDPR rules to UK principals (90-day retention) — same database, different rules per row, no manual sorting. Cross-cutting company-level obligations (breach notification, DPIA filing, DPO requirement) apply as the union of all loaded packs whose principals are present. Full rationale and 24 enumerated edge cases: [ADR-0001](#).

STAKEHOLDERS SERVED

Chief Compliance Officer

PII register ·
Compliance score ·
DPIA · Gap remediation
(per-jurisdiction
breakdown)

General Counsel

Auto DPIA drafting ·
Consent evidence ·
Multi-regulator audit
readiness

Chief Marketing Officer

Per-jurisdiction consent
coverage · Marketing
eligible audience ·
Withdrawal propagation

Chief Financial Officer

Risk score · Penalty
exposure across
regulators · Compliance
ROI · Gap prioritization

PERSONA VIEWS — DASHBOARD + SCOPED GENIE, UC-ENFORCED PER PERSONA

UC-ENFORCED

CCO Persona

CCO View (6 pages,
jurisdiction-aware) +
CCO Genie space

Scope:

personal_data_register ·
pii_findings ·
compliance_gaps ·
discovered_tables ·
consent_coverage_summary

UC-ENFORCED

GC Persona

GC View (5 pages,
jurisdiction-aware) +
GC Genie space

Scope:

compliance_gaps
(critical/high) ·
consent_events_log ·
notice_versions ·
dsr_requests ·
dpi_a_runs

UC-ENFORCED

CMO Persona

CMO View (3 pages, incl.
Consent & Audience) +
CMO Genie space

Scope:

marketing_eligible_principals
· consent_events_log

UC-ENFORCED

CFO Persona

CFO View (4 pages,
incl. Penalty
Exposure) + CFO
Genie space

Scope:

compliance_gaps ·
discovered_tables
(penalty-weighted,
per-regulator)

Each persona user has CAN_READ only on their dashboard, CAN_RUN only on their Genie space, and UC SELECT only on their domain tables. Cross-persona access is blocked at the query layer, not the UI layer. Persona scoping composes with per-data-subject jurisdiction routing — a CMO sees marketing-eligible audiences across loaded jurisdictions, each row evaluated under its principal's pack.

SHARED INTERFACES (DEPLOYER / ADMIN)

Master AI/BI Dashboard + Genie

10-page dashboard with full data estate visibility · Risk/compliance scores · PII inventory · Compliance gaps · Per-jurisdiction drilldown · NL queries via Genie

Consent API

Consented-only audience views · Real-time eligibility checks · `has_active_consent()` UDF · Marketing eligible principals (per-jurisdiction)

Compliance Q&A (Unscoped)

Admin-only full-estate queries · "What critical PII do we hold under DPDP vs UK GDPR?" · Context-aware routing across all tables and packs



MULTI-REGULATION COMPLIANCE MODULES

PII Discovery & Data Inventory

Universal patterns + every loaded pack's region-specific patterns
Sensitivity tiering · UC tags · Per-finding pack attribution
Pattern union catches Aadhaar, NHS Number, SSN, Personalausweis simultaneously
Living personal data register, jurisdiction-aware

Consent Intelligence Engine

Per-pack purpose taxonomy + lawful-basis semantics
Immutable Delta log with CDF · Withdrawal propagation
DPDP §8: opt-in default · UK GDPR Art. 7: granular per-purpose
Notice version tracking per locale + per pack

Compliance Audit & Gap Analysis

Rules from every loaded pack run against the estate
Each gap tagged with its source pack & rule citation
Penalty-weighted risk per regulator (₹ for DPBI · £ for ICO · €/‰ turnover for EDPB)
Remediation tracker, jurisdiction-faceted



REGULATION PACKS — PLUGGABLE DATA, ROUTED PER PRINCIPAL

DPDP India 2023

ACTIVE

9 rules · 5 IN-specific PII patterns (Aadhaar, PAN, IFSC, India phone, India passport)
DPBI authority · ₹250cr/150cr/50cr/5cr penalty ceilings
10 Indian languages · 730-day default retention

UK GDPR

PLANNED · M2

UK GDPR articles · UK-specific patterns (NI Number, NHS Number, postcode)
ICO authority · £17.5M / 4% turnover ceilings
en-GB + cy locales · 90-day default retention

EU GDPR

FUTURE

Articles 5-22, 25, 30, 32-36 · 24 EU official locales
EDPB + national supervisors · €20M / 4% turnover
Drop-in 9-yaml-file authoring per pack contract

CCPA / PIPEDA / LGPD / APPI

FUTURE

Each regulation = one directory under `regulations/<code>/`
No core code change to add a pack
Customer's `regulations/` directory is the deployment knob

Per-data-subject routing — the core architectural primitive.

- | | | |
|-----------------------------------|---|---|
| <code>jurisdiction = 'IN'</code> | → | DPDP rules · 730-day retention · DPDP §11/§12 DSR rights · 10 Indian-language notices |
| <code>jurisdiction = 'GB'</code> | → | UK GDPR rules · 90-day retention · UK GDPR Art. 15-22 DSR rights · en-GB notices |
| <code>jurisdiction = 'EU'</code> | → | EU GDPR rules · per-Member-State retention · Art. 15-22 + Art. 7 lawful-basis · 24-locale notices |
| <code>jurisdiction IS NULL</code> | → | High-severity compliance gap · "Unmapped principals" tile on CCO dashboard |

Cross-cutting company-level rules (breach notification, DPIA filing, DPO requirement, Significant Data Fiduciary registration, audit-log retention) apply as the **union** of every pack whose principals are present in the deployment — a breach affecting both IN and GB principals notifies both DPBI and ICO in parallel.

AGENT BRICKS — AI-POWERED COMPLIANCE

DPIA Auto-Generator

AGENT

Reads UC metadata + pii_findings + compliance_gaps + consent coverage → structured DPIA draft in 30 seconds. Cites every regulation whose principals are in the activity's data scope (DPDP §... AND UK GDPR Art. 35 in the same document).
Replaces 3-month manual exercise · Quarterly cron + Review App

Compliance Q&A

AGENT

Natural language queries against the compliance estate · Context-aware routing across PII, consent, gap, and pack tables · Foundation model: databricks-gpt-oss-120b · Persona text_instructions composed from loaded packs

PII Classifier

AI FUNCTIONS

ai_classify for unstructured fields (diagnosis, prescriptions, notes) · ai_extract for structured PII from free text · Pattern union from all loaded packs catches what regex cannot

GOVERNANCE SPINE

Unity Catalog — Five Enforcement Layers

Discovery: 36 PII column tags · auto lineage (Bronze → Silver → Gold → Register) · sensitivity tiers · foreign catalogs for Lakehouse Federation

Per-persona grants: each persona user has USE_CATALOG + USE_SCHEMA + SELECT only on scoped tables; out-of-scope queries fail PERMISSION_DENIED regardless of UI path

Column masks (DPDP §5(2) · UK GDPR Art. 32): 37 PII columns wrapped in CASE WHEN is_member('admins') THEN val ELSE mask(val) END; persona queries return redacted values

Per-data-subject jurisdiction routing: rule evaluation, retention enforcement, and notice rendering route through customers_tagged.jurisdiction → the pack governing that principal

Audit trail (DPDP §18 · UK GDPR Art. 30): system.access.audit captures every query with user + timestamp; CCO Genie space surfaces it via sample questions

LAKEHOUSE — DELTA MEDALLION

Bronze PIPELINE-ONLY

5 Auto Loader tables (21.5K rows) + 3 Lakeflow Connect tables + 2 federation_mock tables
All STRING · CDF enabled
Zero human grants — DLT pipeline + seed scripts

Silver MASKED

8 typed tables + 2 federation views + 3 metadata
37 PII columns UC-masked
jurisdiction column on customer-level tables
pii_findings · compliance_gaps · access_patterns

Gold

marketing_eligible_principals
consent_coverage_summary
persona_overview_metrics
sensitive_data_exposure · data_lineage

Lakebase (Production)

Consent OLTP tier
Sub-second read-after-write
Delta in POC · Lakebase at scale

PROCESSING & DEPLOYMENT

Lakeflow Declarative Pipelines (DLT)

Bronze → Silver medallion · PII classification (every loaded pack's patterns) · **16 DQ expectations across 5 silver tables** (5 expect_or_drop for primary keys, 11 expect warn for PII format/enum rules) · metrics auto-emitted to DLT event log · *only writer to Bronze and Silver*

Databricks Workflow Jobs

run_medallion (DLT trigger) · phase1_bootstrap (multi-pack rule + gap derivation, jurisdiction-routed · 1K consent events · views/UDFs · idempotent · seed=42) · dpia_generator (quarterly cron, multi-regulator citations)

Databricks Asset Bundle + GitHub Actions

Full IaC via databricks bundle deploy · CI validates on every push/PR: bundle validate + py-syntax + HTML tag-balance + persona-boundary test

DATABRICKS-NATIVE INGESTION LAYER

DATA SOURCE INGESTION

Lakehouse Federation

Databases: Postgres · MySQL · SQL Server · Snowflake · BigQuery
CREATE FOREIGN CATALOG USING CONNECTION
Query in-place · No data movement · UC-governed

Lakeflow Connect

SaaS: Salesforce · Workday · ServiceNow · SharePoint · MongoDB
Managed ingestion · Schema evolution · CDC built-in

Auto Loader + Volumes

Files: S3 · ADLS · GCS · CSV/JSON/Parquet
cloudFiles format · Incremental · Exactly-once

CLIENT SOURCE SYSTEMS

CRM

Salesforce · HubSpot

HRMS

Workday · SAP

Databases

Postgres · MySQL · Oracle

Cloud Storage

S3 · ADLS · GCS

SaaS Apps

ServiceNow · Zendesk

Databricks Platform Features Used

FEATURE	HOW IT'S USED	WHAT IT DELIVERS
Unity Catalog	PII column tags · Auto lineage · Foreign catalogs · Sensitivity tiers · Per-persona grants + column masks (37 cols) + bronze-is-pipeline-only fence + row filters (residency + CMO persona purpose-scope) + per-data-subject jurisdiction routing enforcing CCO/GC/CMO/CFO boundaries at query time	One governance boundary across the estate — an SA or regulator gets an answerable question in a single query instead of navigating a patchwork of app-level checks. Enforcement is at query time, so no UI path can leak what UC has blocked.

FEATURE	HOW IT'S USED	WHAT IT DELIVERS
Regulation packs (the central abstraction)	Every regulation-specific value (rules, notices, retention, residency, region PII patterns, DSR SLAs, DPIA template) lives in <code>regulations/</code> . <code>governance_core/</code> hosts the regulation-agnostic loader + resolver. All packs in <code>regulations/</code> load simultaneously; each customer-level row's jurisdiction column routes rule evaluation per-row. Adding UK GDPR / EU GDPR / CCPA / PIPEDA / LGPD is authoring 9 yaml files plus a <code>pii_patterns.py</code> + <code>dpia_template.yaml</code> — zero core code change.	Same lakehouse + same personas + same dashboards, multiple regulations active per row. A multinational customer applies DPDP to Indian principals and UK GDPR to UK principals from the same query, no per-deployment forking. See ADR-0001 for the per-data-subject routing decision and 24 enumerated edge cases.
system.access.audit	Every SQL query + dashboard view + Genie question captured with user + timestamp (DPDP §18 · UK GDPR Art. 30 evidence). CCO Genie space surfaces it via sample questions.	Multi-regulator audit evidence available on demand. <i>"Who accessed patient X's record last week?"</i> answered live in seconds — same query satisfies a DPBI inquiry and an ICO inquiry.
Lakeview Dashboards + Genie (persona-sliced)	4 persona-specific dashboards sliced from the master (idempotent PATCH on redeploy) + 4 Genie spaces scoped to that persona's <code>data_sources</code> . Persona <code>text_instructions</code> auto-composed from loaded packs. Dashboards published with <code>embed_credentials=false</code> so queries run as the viewer and UC enforces.	Every executive (CCO/GC/CMO/CFO) gets a personal view tailored to their questions, with per-jurisdiction drilldown — not one firehose dashboard. Natural-language agents collapse <i>"file a ticket, wait two days, get a spreadsheet"</i> into a five-second Genie answer that already knows which regulation governs the principal you're asking about.
Lakeflow Declarative Pipelines	Bronze → Silver medallion with 16 DQ expectations · PII classification pipeline running every loaded pack's patterns · metrics to event log	Data quality becomes an always-on control, not a periodic review. The 16 expectations auto-emit pass/fail metrics to the event log, so a DPO can prove "our controls ran every night and caught these rows" without custom observability code.
Agent Bricks + AI Functions	DPIA auto-generation (multi-regulator citations) · Compliance Q&A · <code>ai_classify/ai_extract</code> for unstructured PII	DPIA production drops from ~3 months of legal + IT + compliance coordination to ~30 seconds of agent output + ~1-2 hours of human review. A processing activity touching IN and GB principals produces a single DPIA citing both DPDP §... and UK GDPR Art. 35 inline.

FEATURE	HOW IT'S USED	WHAT IT DELIVERS
		For an enterprise running 20+ DPIAs a year across multiple regulators, the savings compound.
Lakehouse Federation	Foreign catalogs for Postgres, Snowflake, BigQuery — query in place, no ETL	Data residency stays compliant without engineering work — the source system never has to move its data. Cross-border concerns under DPDP §16, UK GDPR Chapter V, GDPR Art. 44-50 — query where the data already lives, residency filter applied at query time.
Lakeflow Connect	Managed ingestion for Salesforce, Workday, ServiceNow, SharePoint	Onboarding a new SaaS source goes from a 2-sprint engineering project to a configuration step. Schema drift, OAuth renewals, pagination and rate-limit handling are platform responsibilities — ingestion stops being a product your team maintains.
Auto Loader	Incremental file ingestion from S3/ADLS/GCS with schema evolution	Exactly-once file ingestion without your team tracking what's been processed. Schema evolution is automatic — partner uploads a new column, your pipeline keeps running instead of paging an engineer at 2am.
AI/BI Dashboard + Genie	10-page compliance dashboard with per-jurisdiction filter · 4 persona-scoped Genie spaces with pack-aware instructions	Compliance insight accessible to business users without SQL literacy. A CMO asks <i>"can I email this customer?"</i> and Genie checks the principal's jurisdiction, applies that pack's consent rule, and returns a yes/no — no analyst ticket, no wait, no raw-PII exposure.
Delta Lake (CDF)	Immutable consent log · Withdrawal propagation · Time travel for audit	Real-time consent withdrawal propagation across regulators — DPDP §8(7) "without undue delay" and UK GDPR Art. 7(3) "as easily as it was given" both satisfied by the platform, not by a nightly batch job. Downstream marketing / ML systems auto-invalidate the moment a principal

FEATURE	HOW IT'S USED	WHAT IT DELIVERS
		withdraws; no polling, no stale cache.
Delta Sharing	Read-only audit share — 3 objects: <code>personal_data_register</code> , <code>compliance_gaps</code> , <code>notice_versions</code>	External auditors query compliance evidence from their own workspace — no local accounts to provision, no VPN, no orphan credentials to clean up after the audit. Revoke in one command when the engagement ends. Same share serves DPBI and ICO auditors.
Databricks Asset Bundle	Full platform as IaC — one command deploy, one command destroy	Reproducibility becomes free. A teammate stands up the entire compliance stack on a fresh workspace in minutes instead of days; every environment looks identical; drift between dev and prod stops existing as a category. Per-customer regulation set is driven by the contents of <code>regulations/</code> , not by environment configuration.
Persona Orchestrator (<code>scripts/setup_all_personas.py</code>)	One command stands up the full persona layer: slice → Genie → attach banners → SCIM users → UC grants → warehouse/dashboard/Genie ACLs. Auto-detects workspace/warehouse/deployer. Idempotent end-to-end. Genie text_instructions composed from loaded packs.	Governance is deployed the same way code is — scriptable, reviewable, reproducible. Onboarding a new persona (e.g. a Chief Risk Officer) becomes a code change with a PR, not a week of click-ops that quietly drifts between environments.
GitHub Actions + persona_boundary_test	On every push/PR: <code>bundle validate</code> + Python AST parse + HTML tag-balance + live 4/4 persona-boundary check against the workspace	Governance is continuously proven, not annually audited. If a future change breaks a persona boundary, CI fails the PR before it merges — regulator-grade enforcement with a commit-level feedback loop.
Lakebase (Production)	Consent OLTP tier with sub-second writes and read-after-write consistency	Consent capture latency goes from "acceptable" to "imperceptible" — a principal withdrawing on your mobile app sees the new state reflected across downstream systems in under a second. One Databricks surface handles both the OLTP write and the lakehouse analytics.

Note on numbers. Counts shown in the module cards and metric tiles (36 PII findings across 10 silver objects, 37 column masks, 135 base compliance gaps from the DPDP-only seed, 1,000 consent events, 292 principals, 9 DPDP rules) are **illustrative** — they come from the synthetic POC dataset on a Databricks Free Edition workspace (seed=42) and are reproducible via the SQL in [how_to_test.html](#). With UK GDPR and additional packs loaded the absolute counts grow (more PII patterns hit, more rules fire); the transferable signal is the shape of the ratios (critical-to-total gap mix, auto-classify rate, consent-grant rate per jurisdiction), not the specific numbers.

Compliance Pack Accelerator · Databricks-Native Multi-Jurisdiction Architecture · May 2026 · [ADR-0001 governs this design](#)